The Well Runs Dry

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Introduction

Kelvin Sampson has enacted a complete turnaround of the Houston Cougars program. After the days of Drexler and Olajuwon, the Cougars reached the NCAA Tournament just four times in thirty years; all of those berths ended in first round exits. Since Sampson became head coach in the 2014-15 season, Houston has made the tournament eight times. Their last seven trips reached at least the Sweet Sixteen, and two campaigns reached the Final Four.

Houston has become one of the most respected programs in America. They have won 30 games in five consecutive seasons and they have long outgrown their dismissive label as a mid-major, especially since they joined the Big 12 prior to the 2023-24 season. They went 19-1 in conference play in their second season in the scariest conference in the country. They have earned the reputation of making their opponents' lives as miserable as possible. Sampson gets all of his players to buy in completely and what transpires is a relentless style of play. Facing off against Houston is the equivalent of a forty-minute torture chamber.

For a brief snapshot of Houston's remarkable consistency, here's a table displaying Houston's records and team ratings since the 2018 season, when they became a ranked team who made the tournament every year. This is when Houston became the unit that we know today.

Sustained Excellence			NCAA Stat Percentile		
Season	Wins	Losses	ORtg	DRtg	Net
2017-18	27	8	90.3	96.6	97.4
2018-19	33	4	91.8	98.9	99.2
2019-20	23	8	85.8	92.4	93.5
2020-21	28	4	98.8	98.6	99.7
2021-22	32	6	96.6	99.2	99.7
2022-23	33	4	98.9	99.4	100.0
2023-24	32	5	90.9	100.0	99.4
2024-25	35	5	94.5	99.5	99.5
2025-26	30	7	90.7	98.4	97.8

Figure 1: The Cougars' Percentiles of Off., Def., and Net Rating

Despite their regular season dominance and mainstay status in March Madness, Sampson's squad has not gotten over the hump. Their trips in March have all fallen

short of a title. Houston has seen many groups, players, and stars come and go, but none have broken through. At age 70, retirement nears for Kelvin Sampson, especially in the exhausting age of NIL and the transfer portal. He already announced his return for the 2026-27 season, so he'll have another crack at building a contending unit. However, some great players and teams have taken the floor in Sampson's time at Houston, so how is it that they haven't won an NCAA championship yet?

The Brutality of March

Here's a snapshot of Houston's yearly finishes in March Madness, once again, since 2018.

Season	Seed	Result	Score	Opponent
2017-18	6	L, R32	64-63	(3) Michigan
2018-19	3	L, S16	62-58	(2) Kentucky
2019-20	*COVID*			
2020-21	2	L, F4	78-59	(1) Baylor
2021-22	5	L, E8	50-44	(2) Villanova
2022-23	1	L, S16	89-75	(5) Miami
2023-24	1	L, S16	54-51	(4) Duke
2024-25	1	L, Final	65-63	(1) Florida
2025-26	2	L, S16	65-55	(3) Illinois

Figure 2: Houston's March Résumé

Hidden within these scores are pain, suffering, and heartbreak. I can't entirely communicate the level of misery the Houston faithful have been subjected to, but here's a rap sheet of those losses in grim detail:

Quick couple of formulas:

Effective Field Goal Percentage ($eFG\%$) and True Shooting Percentage ($TS\%$)

$$eFG\% = (FG + 0.5 * 3P)/FGA$$
$$TS\% = PTS/(2 * (FGA + 0.475 * FTA))$$

- 2018
  - Jordan Poole buzzer beater
  - Houston had 8 assists to 10 turnovers
  - Devin Davis missed two free throws up 63-61 with 3.9 left (he was 9-10 prior to the two misses!)
  - That Michigan team did reach the title game
- 2019

  - They were down 39-26 early in the second half, the defense tightened up and Armoni Brooks caught fire to bring them back.
  - They took a 58-55 lead with less than a minute left, and then they didn't score again.
  - This Kentucky team was loaded (6 NBA players)
  - Houston was out-rebounded 34-21
  - They were also out-shot (.462 vs .522 $eFG\%$), (.494 vs .563 $TS\%$)
  - They committed far more fouls (23 to 13)
- 2021
  - The Cougars were down 25 at halftime.
  - They just couldn't score (.436 vs .627 $eFG\%$), (.471 vs .638 $TS\%$)
  - Interesting tournament run, the Cougars played four straight teams with a double-digit seed (15, 10, 11, 12)[1]
  - That Baylor team was a machine (28-2); they beat Gonzaga in the title game.
- 2022
  - Houston was down 40-29 in the second half; they did cut the deficit to two.
  - 29.8 $FG\%$, 1-20 from downtown[2]
  - Three key players had brutal shooting days.
  - Kyler Edwards (1-12, 0-8 $3P$)
  - Taze Moore (6-21)
  - Jamal Shead (4-13)
  - Houston was missing Marcus Sasser (injured early in season), and it showed.
- 2023
  - Out-shot very badly (.445 vs .608 $eFG\%$), (.510 vs .645 $TS\%$)
  - That Canes team went to the Final Four
  - Tough day for freshman Jarace Walker (4-16)
- 2024
  - Jamal Shead got hurt early; the offense suffered in his absence (10 assists to 10 turnovers)
  - Shead was a first-team All-American
  - The Cougars were up six near the end of the first half when Shead left with his ankle injury.
  - Poor shooting once again: 2-8 $3P$, 9-17 FT
- 2025

[1] The 12 seed was Oregon State, a very impressive team led by Ethan Thompson, who very much outplayed their 12-seed.

[2] A Houston team going ice cold from three in a win-or-go-home game? No way.

  - We all remember the ugly ending, they didn't get a shot off in the final possession.
  - They didn't get a shot attempt during the previous possession either.
  - Houston led 42-30 with about fifteen minutes left.
  - Florida led for a grand total of 63 seconds, meaning $\frac{1}{40}$th of the entire game.
  - Houston had an unreal comeback against Duke in the previous round
  - Unlike 2021, the Cougars had an extremely tough road in pursuit of a title (beating "8 seed" Gonzaga, (4) Purdue, (2) Tennessee, (1) Duke, (1) Florida)
  - Houston really struggled to score (.391 vs .453 $eFG\%$), (.416 vs .516 $TS\%$)
  - Cryer (6-18), Roberts (3-13), Uzan (2-9), Sharp (3-11)
  - Truly a nightmare game for Emanuel Sharp; he had two huge turnovers at the end. There wasn't a shot he didn't like, until the end of the game when his last-second pump fake created the painful moment.
  - Arguably the best Houston team in the Sampson era, despite Shead's departure

- 2026
  - 2nd half began with Houston down two, Illini went on a 17-0 run, Houston didn't score for about SEVEN minutes
  - Guess what, they shot poorly again: (.414 vs .520 $eFG\%$), (.423 vs .533 $TS\%$)
  - 2 free throw attempts to Illinois's 21

See the theme? This is eight tournament runs that ended, more or less, because Houston's offense dried up. On the surface, this is confusing, since their offensive rating has been in the top ten percentile for eight out of the nine seasons since 2018 (see Figure 1). However, that number requires some context, and I'm going to explore the parts of Houston's offense that are not elite. The Cougars' offense is very strong by offensive rating, but they lack consistent half-court production and efficiency.

No Offense..

Offensive rating must be acknowledged; Houston's offense is very productive. However, a good word for their offense is "opportunistic." As we saw in Figure 1, Houston's calling card is their incredible defense. They use relentless pressure to bully opponents into submission. Their defense is elite and their opponents are all but guaranteed to have a poor shooting night.

The problem is that their offense is not potent on its own; Houston is elite in two areas: Offensive rebounding and limiting turnovers. Combined with their defense, Houston's game is generating extra possessions, out-rebounding, and attempting more shots than their opponent.

Suffocating	Percentile	
season	DRtg	Opp eFG%
2017-18	96.6	98.6
2018-19	98.9	100.0
2019-20	92.4	99.2
2020-21	98.6	100.0
2021-22	99.2	99.2
2022-23	99.4	99.7
2023-24	100.0	99.7
2024-25	99.5	99.5
2025-26	98.4	96.2

Figure 3: Houston's Def. Rating and Opponent $eFG\%$ percentiles

Control	Percentile	
Season	OREB%	TOV%
2017-18	88.6	78.1
2018-19	91.8	93.2
2019-20	99.7	86.7
2020-21	99.7	98.3
2021-22	99.7	88.0
2022-23	99.2	97.0
2023-24	97.0	98.9
2024-25	96.7	97.5
2025-26	89.0	99.5

Figure 4: Houston's rankings in Offensive Rebounding Rate and Turnover Rate.

Okay, so we know a few things: Houston's defense makes it incredibly challenging for their opponent to score, their offensive rebounding ability generates extra possessions, and they limit turnovers, which limits extra possessions and/or fast break opportunities for the opposition. What we haven't yet seen is anything that describes how well they actually put the ball in the basket.

Elite?	Percentile		
Season	ORtg	eFG%	TS%
2017-18	90.3	76.1	75.2
2018-19	91.8	66.0	65.7
2019-20	85.8	22.9	30.9
2020-21	98.8	62.0	62.0
2021-22	96.6	84.6	71.5
2022-23	98.9	72.5	70.0
2023-24	90.9	40.1	34.0
2024-25	94.5	58.0	52.7
2025-26	90.7	51.8	47.7

Figure 5: Houston's rankings in $eFG\%$ and $TS\%$

This explains the array of tournament games where the Cougars went cold; their advanced shooting has never been within the top fifteen percent of the NCAA. That might seem a little arbitrary, but there is something real here. If a team's ability to put the ball in the basket is not incredibly reliable over an entire season, then the well is very likely to run dry in a March Madness game against an elite opponent. Additionally, these mid-tier percentiles stick out further compared to the NCAA teams that actually reached the Final Four.

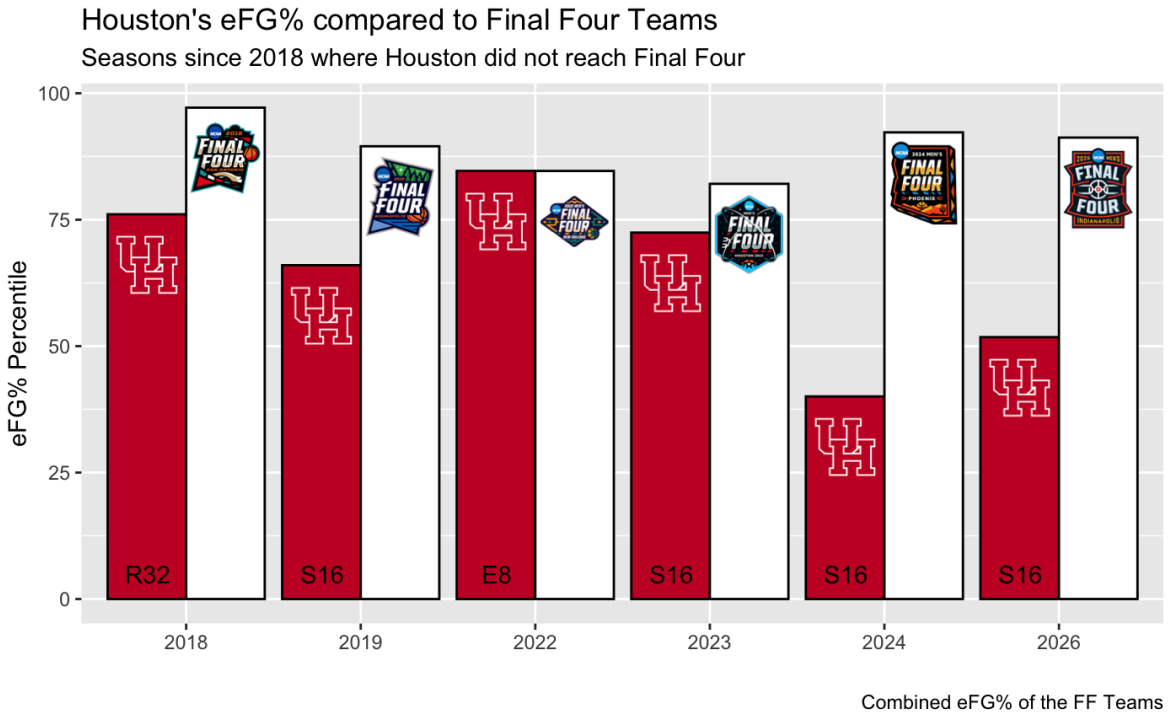

Figure 6: Houston's standing in terms of $eFG\%$ among the Final Four teams

The obvious exceptions are the 2022 and 2023 seasons. Those squads were loaded with several players who could consistently score and/or create their own shot. Other

Houston squads usually have one or two players who are capable of creating their own offense.

Season	Player	Summary
2017-18	Robert Gray Jr.	19.3 pts (led AAC), 35.9 3P%
2017-18	Corey Davis Jr.	13.1 pts, 42.9 3P%
2020-21	DeJon Jarreau	10.6 pts, 34.4 3P%, 4.3 ast
2020-21	Marcus Sasser	13.7 pts, 33.5 3P%
2020-21	Quentin Grimes	17.8 pts, 40.3 3P% (led AAC)
2021-22	Jamal Shead	10.0 pts, 5.8 ast (led AAC)
2021-22	Kyler Edwards	13.8 pts, 33.3 3P%
2021-22	Fabian White, Jr.	12.5 pts, 37.1 3P%
2021-22	Josh Carlton	11.6 pts, 62.6 FG%
2021-22	Taze Moore	10.4 pts, 2.9 ast
2021-22	*Missing Tramon Mark (7 G) and Marcus Sasser (12 G)*	
2022-23	Jamal Shead	10.5 pts, 5.4 ast
2022-23	Marcus Sasser	16.8 pts, 38.4 3P%, 3.1 ast
2022-23	Tramon Mark	10.1 pts
2022-23	Jarace Walker	11.2 pts, 34.7 3P%
2022-23	J'Wan Roberts	10.0 pts, 61.2 FG%

Figure 7: Some of the notable playmakers when Houston had a stronger offense

Even in the seasons in which Houston made the Final Four, their offense was not lighting the world on fire.

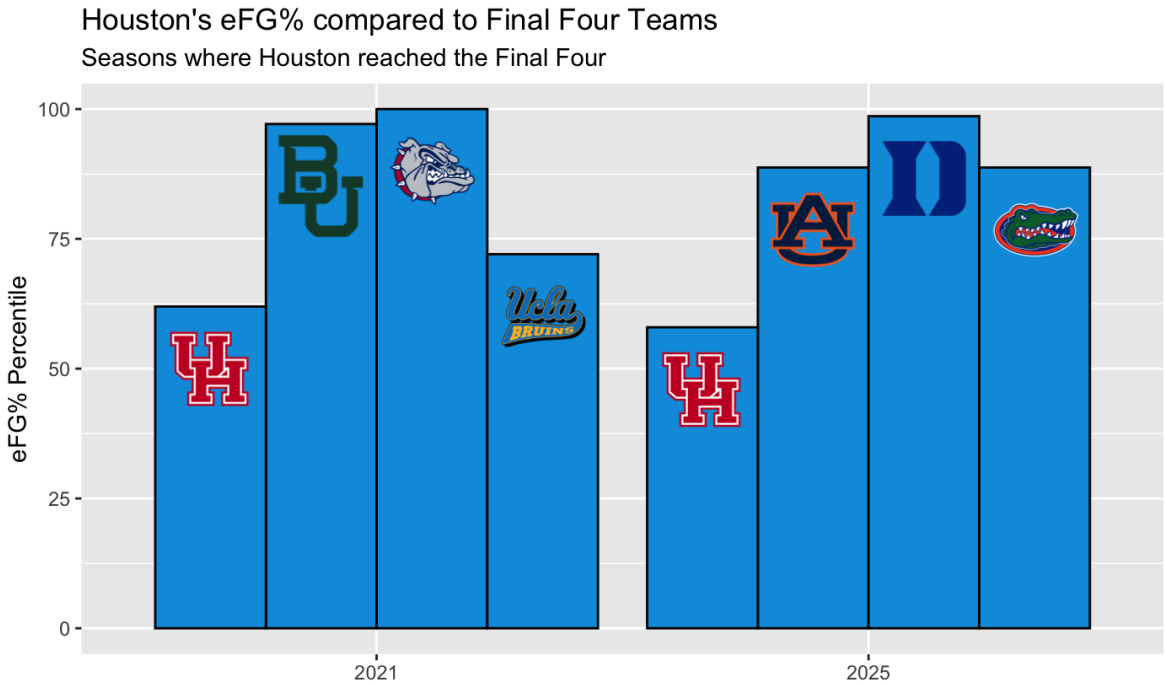

Figure 8: Houston's $eFG\%$ standing in the two Final Four runs.

UCLA being as low as they are is very misleading; that Bruins squad had FOUR players who could all create their own shot: Johnny Juzang, Jaime Jaquez Jr., Jules Bernard, Tyger Campbell. That Bruins team was almost an anti-Houston; they had a below-average defense but a very strong offense. They played at a slower pace than 95% of teams in March and relied on the creation ability of their Big Four to win games. Their team was straight buckets; it would have led to a title game berth if not for Jalen Suggs's heroics.

The 2024 season was the most difficult for Houston. They lost Sasser, Mark, and Walker. In their absence, Roberts and Emanuel Sharp assumed more offensive responsibilities. Sharp is a gunner (36.8 $3P\%$, 48.6 $eFG\%$) with no conscience, and Roberts is an old school post player who uses aggressive power moves to generate shots, but struggles with touch and free throw shooting (51.1 $FT\%$). Sounds like a recipe for inefficient scoring. Jamal Shead had an All-American campaign, but efficient scoring of his own was not his specialty (40.9 $FG\%$, 30.9 $3P\%$) They did add Baylor's LJ Cryer, and he immediately became Houston's safety blanket. He led the Big 12 from distance (38.8 $3P\%$) and was the only Cougar who could consistently and efficiently create his own shot. Roberts had a solid post game, but he largely benefited from the playmaking of Jamal Shead.

When changing the metric to true shooting, those graphs paint a very similar picture, but the difference is more extreme.

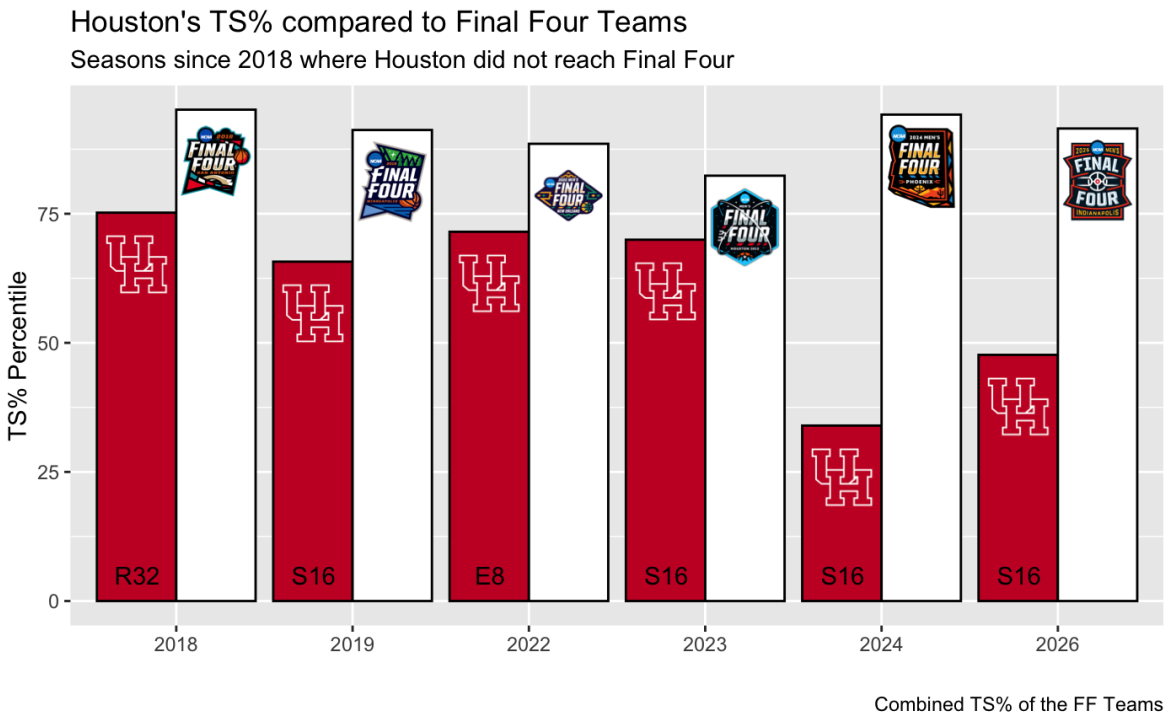

Figure 9: Houston's standing in terms of $TS\%$ among the Final Four teams.

This is the crux of the problem. Their offense is not very self-sufficient. Houston relies on defense powering their offense. In the halfcourt, they rely on limiting turnovers and generating extra possessions with offensive rebounds. That style of play leads to a slow, grinded style of play, where Houston typically has more rebounds and more shot attempts. This play style has failed against big teams that can execute in the half court (Florida, Illinois, Baylor, Miami, Kentucky).

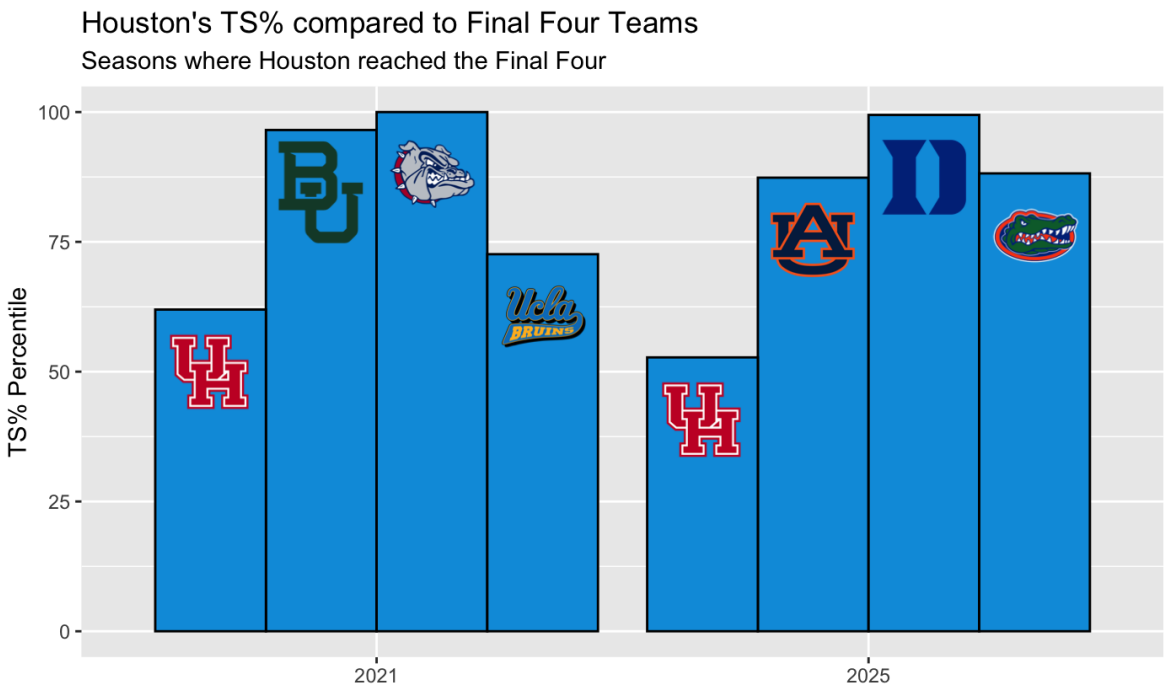

Figure 10: Houston's standing in terms of $TS\%$ when they made the Final Four.

They don't have stars/go-to guys on the same level as teams they've encountered. Players like Gray, Grimes, Sasser, Cryer were able to create their own offense, but it wasn't always very efficient. They have had two elite playmaking guards in Shead and Jarreau who were capable of scoring. They had streaky, gritty players who could contribute, even if it was ugly at times, like Roberts and Mark. This past year, Houston had perhaps their most talented offensive player in the Sampson era: Kingston Flemings. And yet, he was not very aggressive in tournament play. The 2026 Cougars had to rely on Sharp, Chris Cenac, and Milos Uzan having a down year. Cenac and Sharp performed admirably, but they just could not provide enough consistent offense.

More On The Gap

It's been established that a large gap has formed between Houston and the top dogs of March Madness in terms of shooting efficiency. 2022-23 was the last season that Houston had a roster full of weapons, and since then, they have been ultra-reliant on one or two players (Shead, Roberts, Cryer, Flemings, Sharp). I was curious how their now mediocre shooting fares once March rolls around, and I ran a few different bootstrapping tests to supply further context.

Some notation first:

- θ_C: Statistic θ in conference play and conference tournament
- θ_M: Statistic θ in March Madness

These bootstrapping results are comparing Houston and the Final Four teams from 2018-2026 (2020 excluded).

H_o	Team(s)	LB	UB	$0 \in$ CI	Verdict
$eFG_C - eFG_M = 0$	Houston	0.004544	0.06516	No	**Reject** Ho
$eFG_C - eFG_M = 0$	Final Four Teams	-3.536e-05	0.02611	Yes	Fail to reject H_o
$TS_C - TS_M = 0$	Houston	0.01429	0.04870	No	**Reject** Ho
$TS_C - TS_M = 0$	Final Four Teams	0.005700	0.01988	No	**Reject** H_o

Table 1: Bootstrapping Results Pt. 1

There are a few takeaways from these tests:

1. The second test: With a 95% CI, the $eFG\%$ among the (non-Houston) Final Four Teams narrowly evaded rejection. This means that there was not a statistically significant difference between $eFG\%$ in March vs conference play, while Houston's test resulted in H_o being rejected.

2. Distribution differences: In the $TS\%$ tests, while both null hypotheses were rejected, there was a stark difference in distribution. The median decrease of Houston's interval was about 3.1% compared to about 0.9% in the other Final Four Teams, which is a large difference. In the eFG interval, the median drop is roughly 1.3% for the other Final Four teams compared to approximately 3.5% for Houston.

Naturally, I ran two bootstrapping tests on the difference of differences. I'm going to use superscripts $\theta^{(H)}$ and $\theta^{(FF)}$ to refer to stats for Houston and the other Final Four teams, respectively.

$$H_o : (\theta_C^{(H)} - \theta_M^{(H)}) - (\theta_C^{(FF)} - \theta_M^{(FF)}) = 0$$
$$H_a : (\theta_C^{(H)} - \theta_M^{(H)}) - (\theta_C^{(FF)} - \theta_M^{(FF)}) > 0$$

In plain English: I'm trying to prove Houston's shooting drop off in March is greater than the drop off experienced by the other Final Four teams.

Stat	LB	UB	$0 \in$ CI	Verdict
$eFG\%$	-0.01078	0.05546	Yes	Fail to reject H_o
$TS\%$	0.0005191	0.03769	No	**Reject** H_o

Table 2: Bootstrapping Results Pt. 2

Only one of the tests were significant, but both intervals are centered on a positive number. This lends credence to the notion that the way Houston is constructed leads to greater struggles in March. Their reliance on defense, offensive rebounding, and limiting turnovers works very well against most teams, but it generally won't be enough in the heart of March against America's elite offenses. Their struggles to generate half court offense are reflected in the steeper shooting dips in March Madness.

Not Areas of Concern..?

There are two areas I suspected to be factors in Houston's poor offensive output: Joining the Big 12 and Pace. However, after some analysis, I don't believe either of them are necessarily factors in Houston's shortcomings.

2023-24 was the first season in which Houston joined the Big 12. I wondered if the uptick in competition could explain the dip in shooting efficiency. Perhaps the jump from the AAC to the Big 12 was too steep, which would be plausible. It's not like other Big 12 teams are unable to rank within the top ten percent in $eFG\%$ and $TS\%$. However, two factors prevented me from accepting this line of reasoning. One, the roster turnover between '22 and '23 was so great (only 50% of total minutes and points returned) that comparing stats from both seasons without that context is far too crude. Two, even when that crude comparison was done, the results were not significant.

$$\theta = S_{\text{before Big 12}} - S_{\text{after joining}}$$
$$H_o : \theta = 0$$
$$H_1 : \theta \geq 0$$
$$B = 2000 \text{ iterations with } \alpha = .05$$
$$\text{Reject } H_o \text{ if } 0 \notin [\hat{\theta}_{(.025B)}, \hat{\theta}_{(.975B)}]$$

Stat	Lower Bound	Upper Bound	Verdict
$eFG\%$	-0.01272	0.02074	Fail to Reject
$TS\%$	-0.005741	0.01382	Fail to Reject

Table 3: Cheeky bootstrapping

In addition to lack of consistent shotmaking, I took aim at Houston's pace. They are a very slow and deliberate offense. They do space the floor very well; Sampson always builds a unit that can spread the floor and force the defense into difficult rotations. However, that can fall short in seasons where they don't have a creator who can consistently get two feet in the paint and make good decisions: Jamal Shead being the best example of this. Even with the playmaking of Shead, Houston's offense was still very slow. I began to wonder if their snail-like pace was holding them back. However, what I found did not necessarily suggest that Houston's pace is the trait of a loser (Figure 11).

Houston became a fashionable Saint Mary's in 2022. However, they have company among slower teams. The aforementioned UCLA team is on their level, although the more recent Houston teams are slower. Two teams that beat them are in their neighborhood of pace (2022 Villanova and 2026 Illinois). There are even examples of slow teams winning the Big Dance. The dominant 2024 UConn Huskies team is one example, and the 2019 Virginia Cavaliers who completed their redemption tour. The 2019 Cavs were the second slowest team in the country the year they won their title. However, their glacial pace was accompanied by $eFG\%$ and $TS\%$ within the top ten percent of the country. They pounded the ball for thirty seconds, but it led to points on a very efficient basis.

There are more champions on the faster side of the spectrum; this is intuitive since the teams with the most dynamic offenses are set up to go far in March. The teams on the faster side were all loaded. For example,

- 2026 Michigan (Cason, Cadeau, McKenney, Burnett, Gayle, Yaxel, Morez, Mara)
- 2025 Florida (Clayton, Richard, Aberdeen, Martin, Haugh, Condon, Chinyelu)

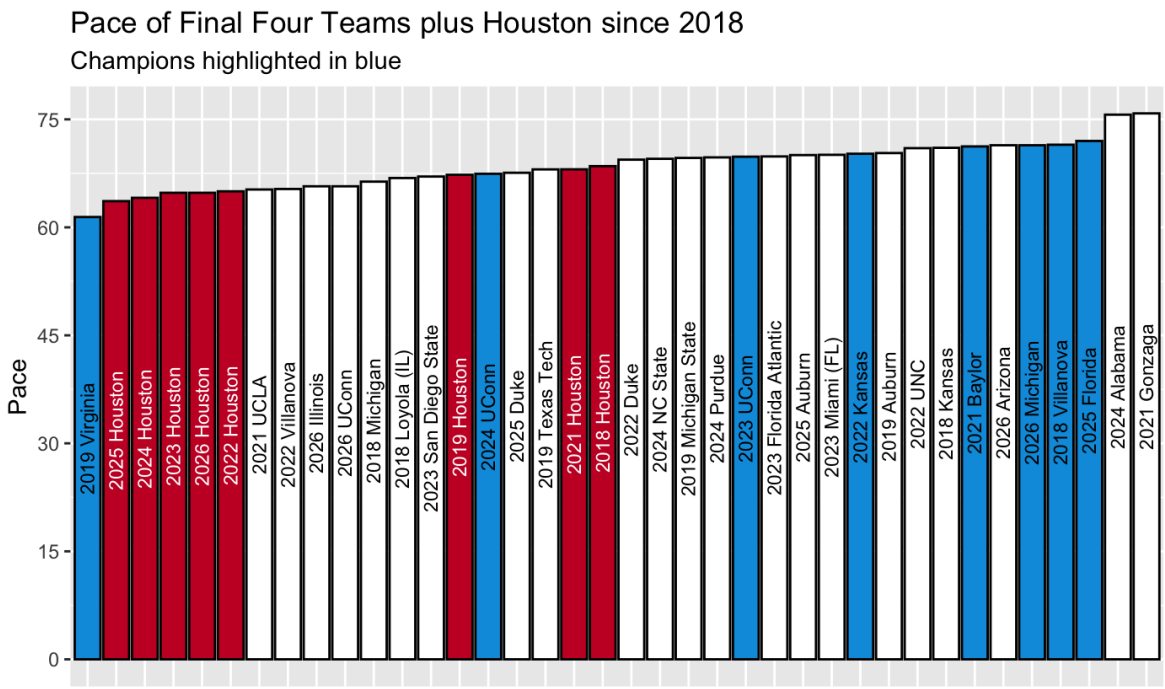

Figure 11: The Pace of Final Four Teams plus Houston teams since 2018

- 2023 UConn (Newton, Hawkins, Jackson, Karaban, Sanogo, Clingan)
- 2022 Kansas (Harris, Agbaji, Braun, Martin, Wilson, McCormack)
- 2021 Baylor (Butler, Mitchell, Flagler, Teague, Mayer)
- 2018 Villanova (Brunson, DiVencenzo, Booth, Bridges, Paschall, Spellman)

[3] This is another manifestation of the gap between Houston and these teams that got over the hump: Talent. Coach Sampson maximizes the units he is given; the teams always play better than the sum of their parts. However, playing above expectations can only catapult Houston so far. It's difficult to win a title without a truly dominant offense and without an array of weapons that can consistently deliver. Houston closes the gap with superior effort, energy, and execution. It's quite remarkable that they pushed that daunting Florida squad to the last possession last year, and it speaks to the level that Coach Sampson is on. Houston's pace is not necessarily a problem, but it's emblematic of their rigid offense and general shortage of offensive talent.

Conclusion

Under Kelvin Sampson, Houston's mantra is very simple: We are going to be tougher, grittier, and better conditioned than you. We will outwork you, we will wear you down, and eventually you will give up. It's as if the late 1990s New York Knicks were in the Big 12. However, this doesn't always work against the elite teams deep in the

[3] Even the slower title teams were scary. 2019 Virginia had Kyle Guy, Ty Jerome and D'Andre Hunter. 2024 UConn returned many of the same players from '23 and added Stephon Castle and Cam Spencer, and they were even more dominant.

tournament; they're battle-tested and have demonstrated that same level of toughness along with a supreme level of talent and offensive execution. Their relentless nature has come up short against teams that are tough, but also more polished and efficient offensively. Their high offensive rating is in large part due to their elite ability to take care of the ball and crash the offensive glass. Their shooting is usually slightly-above average, but it has never been a top-10 percentile unit. Their ability to crash the glass and play clean means they rarely get blown out, but it doesn't guarantee scoring. In March, it creates this image of a team that plays frantically, but it can look very ugly. It's amazing how close some of their March losses were despite being thoroughly outgunned; it's a testament to their second-to-none work ethic and defensive identity. On their worst days, Houston's style doesn't lead to winning, just to slower losing. The most painful examples are the March losses against Illinois, Kentucky, Villanova, and Florida. Without an elite, reliable, self-sufficient offense, a March Madness title is out of the question.

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[10] Statistics are taken from *Sports-Reference*.

[11] Data Analysis, Aggregation, and Manipulation was done using Python (Pandas, BeautifulSoup, PlotNine) and R (tidyverse).

[12] Charts and tables were made with Excel and R ggplot.